

HARMONIC1

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Problem

The incircle of triangle ABC touches BC, CA, AB at points A_1, B_1, C_1 , respectively. The perpendicular from the incenter I to the median from vertex C meets the line A_1B_1 in point K . Prove that CK is parallel to AB .

Solution. Let $C_1I \cap A_1B_1 = J$, N is the intersection of C_1I and the line parallel to AB passing through C . Let $A_1B_1 \cap CI = O$. Since $\angle INC = \angle CIJ = 90^\circ$, we have that $JOCN$ is cyclic. Since $IO \cdot IC = r^2 = IJ \cdot IN$, we know that N is the inverse of J .

Let C'_1 be the antipode of C_1 . Since $IC^2 = r^2 = IJ \cdot IN$ and I is the midpoint of $C_1C'_1$, we know that $(C_1, C'_1; J, N) = -1$. Projecting that line from C to AB , we get $(C_1, L; F, P_\infty) = -1$ where $F = CJ \cap AB$ and $L = CC'_1 \cap AB$. This means that F is the midpoint of LC_1 .

It is well known that L is the ex-touch point of I_A with AB , so, $AL = BC_1$ implying that F is the midpoint of AB .

So, CF, C_1N, A_1B_1 are concurrent at J . Since they are the polars of K, P_∞, C respectively, so, $C - K - P_\infty$. This of course implies that $AB \parallel CK$.